

Synoptic CB: Lateral Transport Campaigns 2024

GWJ Discrete DIC Samples

2025-09-17

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##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE
Date_Run = "7/18/24" #Date that instrument was run
Run_by = "Stephanie J. Wilson" #Instrument user
Script_run_by = "Stephanie J. Wilson" #Code user
run_notes = "The TN data is unusable. TN analyzer broke during run." #any notes from the run
samples <- c("GCW", "LatT") #whatever identifies your samples within the same names
samples_pattern <- paste(samples, collapse = "|")
#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
chks_name = "Chk_Std_1CN" #what did you name your check standards?

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name = "Raw_GCW_Data/GCW_TOCTN.txt"

#file path and name for raw all peaks file
```

```

raw_allpeaks_name = "Raw_GCW_Data/GCW_TOCTN_allpeaks.txt"

#file path and name of processed data file
processed_file_name = "Processed_GCW_Data/GCW_DOC_processed.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
# Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"

#qaqc log file path for this year
Log_path = "Raw_GCW_Data/COMPASS_Synoptic_TOCTN_QAQClog_2025.csv"

```

##Set Up Code

0.1 Import Data Functions

0.2 Import Sample Data

Import Sample Data

New names:

* ' ' -> '...14'

A tibble: 6 x 4

##	sample_name	npoc_raw	tdn_raw	run_datetime
##	<chr>	<dbl>	<dbl>	<chr>
## 1	LatT_GCW_ESTUARY1_LT_20240715_1730	4.66	0.0881	7/19/2024 3:31:23 AM
## 2	LatT_GCW_ESTUARY2_LT_20240715_1730	4.63	0.0775	7/19/2024 3:55:34 AM
## 3	LatT_GCW_ESTUARY3_LT_20240715_1730	4.44	0.416	7/19/2024 4:18:46 AM
## 4	LatT_GCW_UP1-FW1_20240715_1500	1.97	2.19	7/19/2024 4:41:26 AM
## 5	LatT_GCW_UP2-FW1_20240715_1515	1.32	1.26	7/19/2024 5:03:55 AM
## 6	LatT_GCW_TR1_20240715_1530	17.3	37.6	7/19/2024 5:33:14 AM

0.3 Assessing Standard Curves

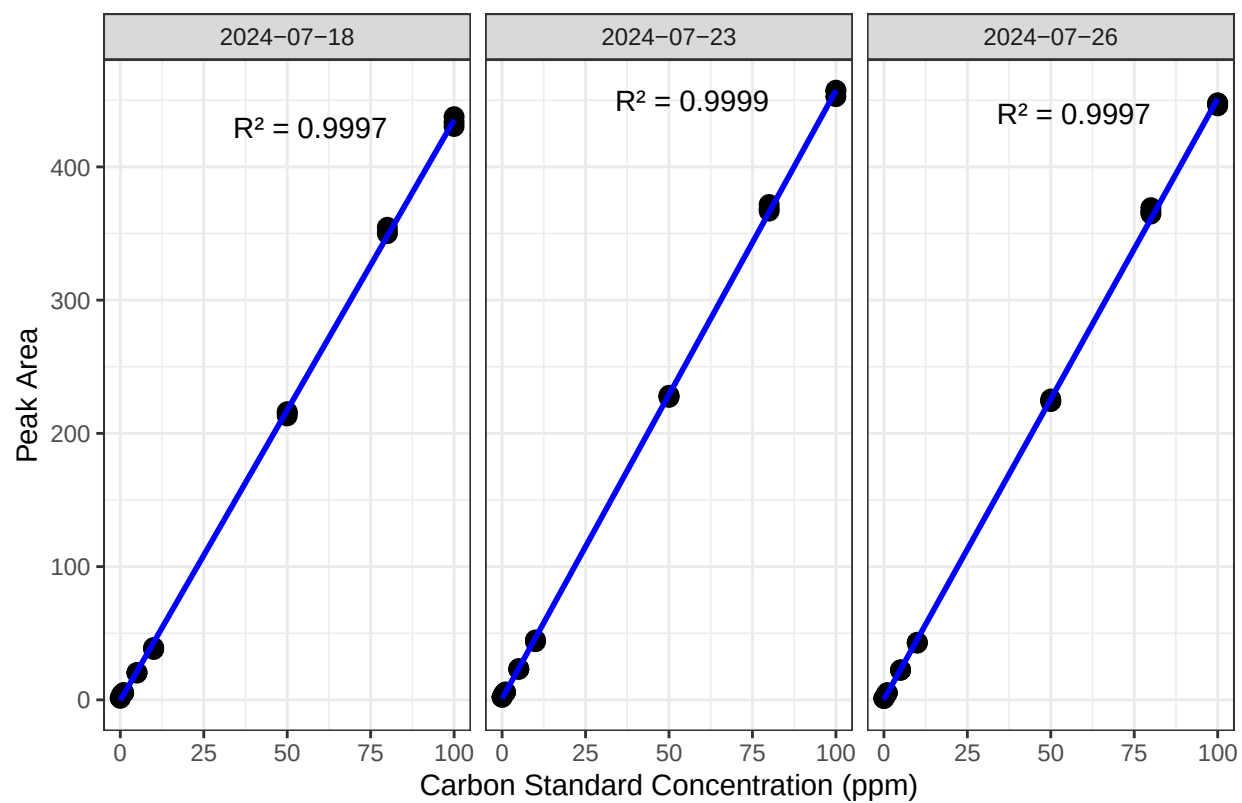
Assess the Standard Curves

New names:

'geom_smooth()' using formula = 'y ~ x'

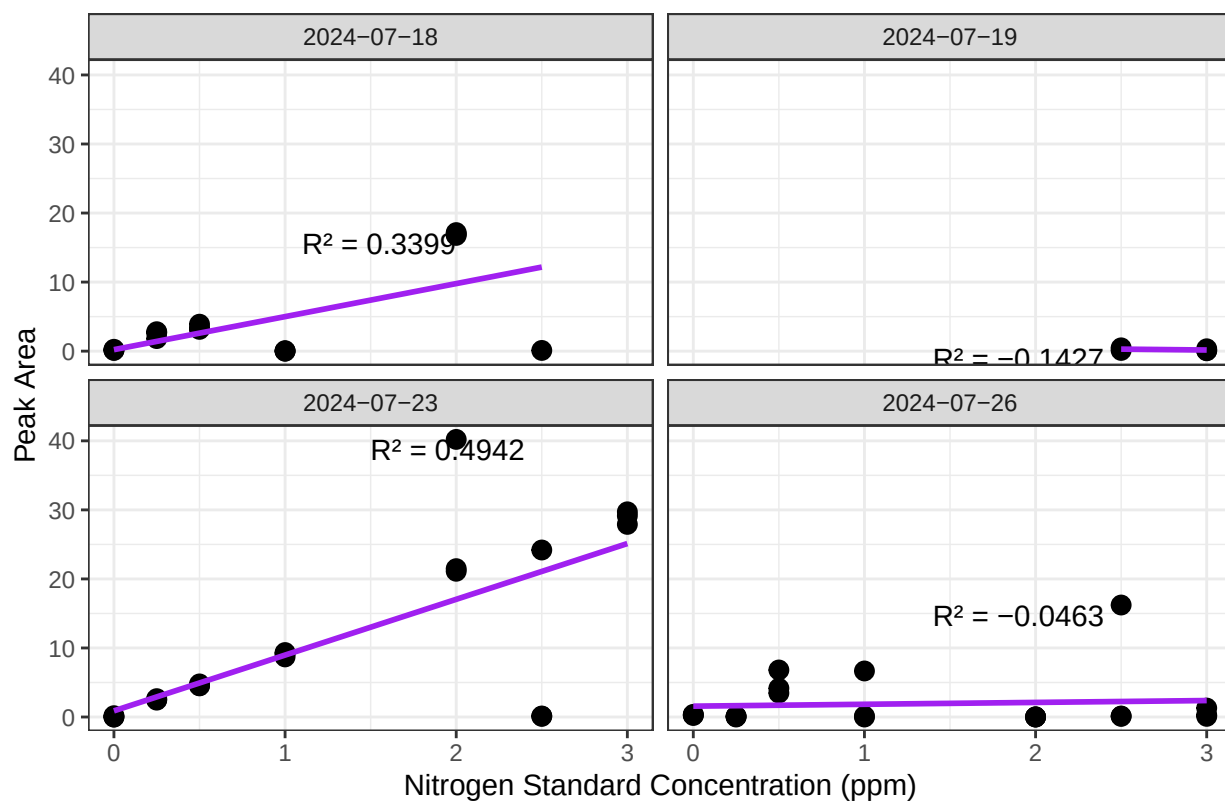
* ' ' -> '...18'

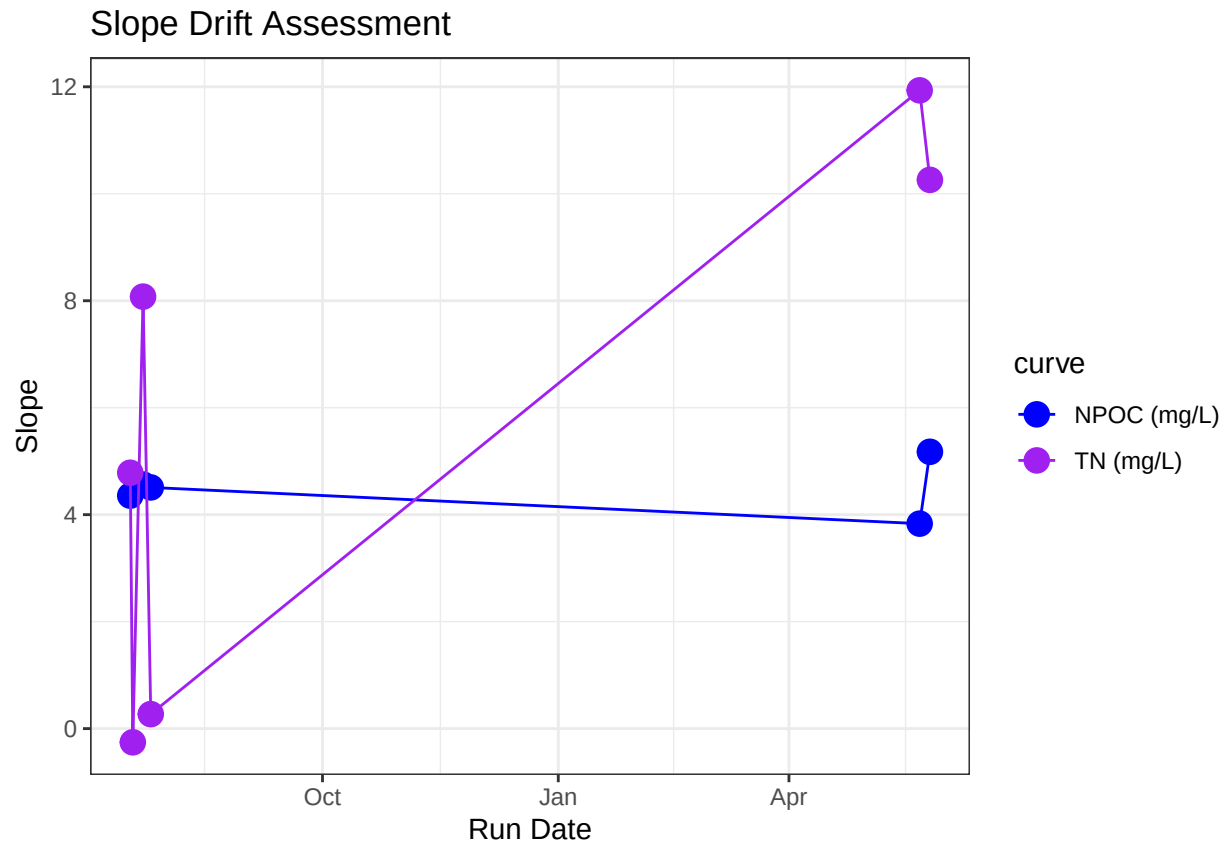
NPOC Std Curve by Date



```
## 'geom_smooth()' using formula = 'y ~ x'
```

TN Std Curve by Date





```
## [1] "NPOC Curve r2 GOOD"
```

```
## [1] "TN Curve r2 is below cutoff! - REASSESS"
```

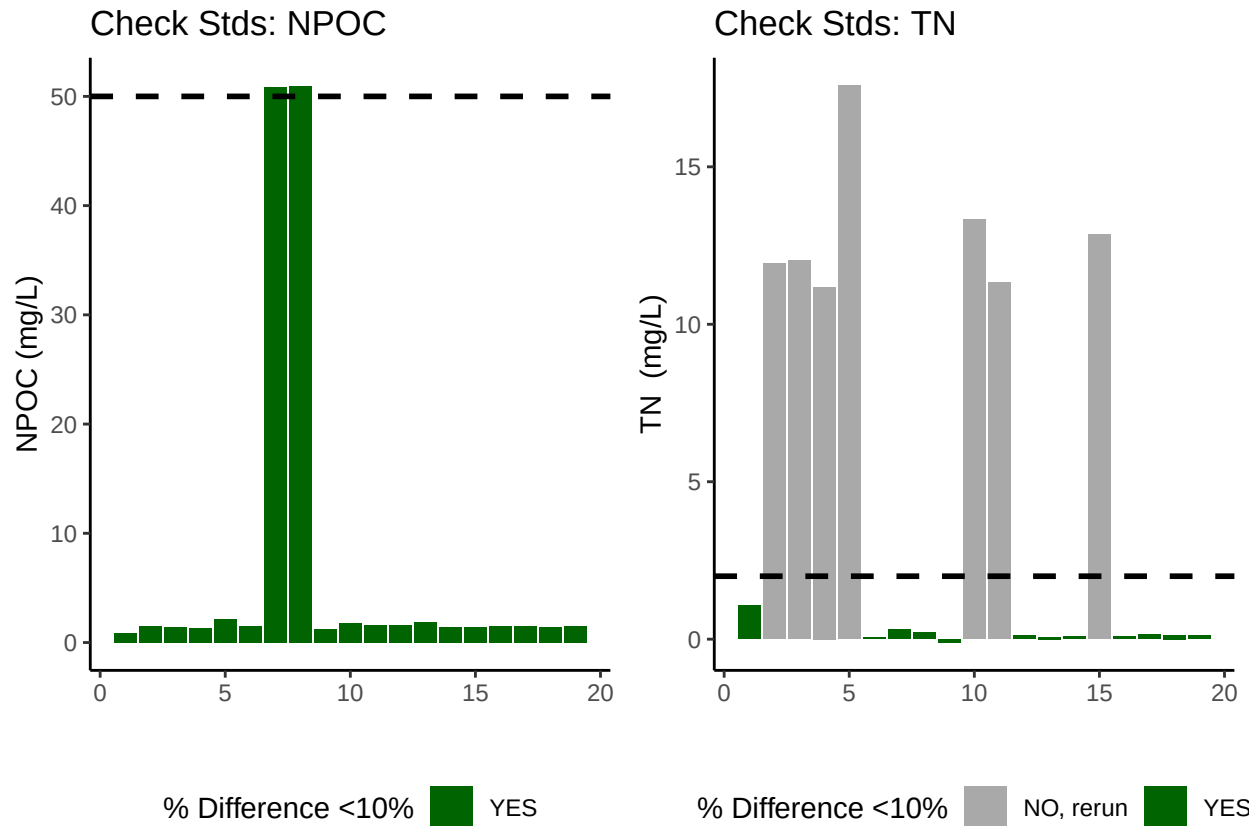
0.4 Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] "Carbon CHECK STANDARD RSD TOO HIGH - REASSESS"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

0.5 Assess Blanks

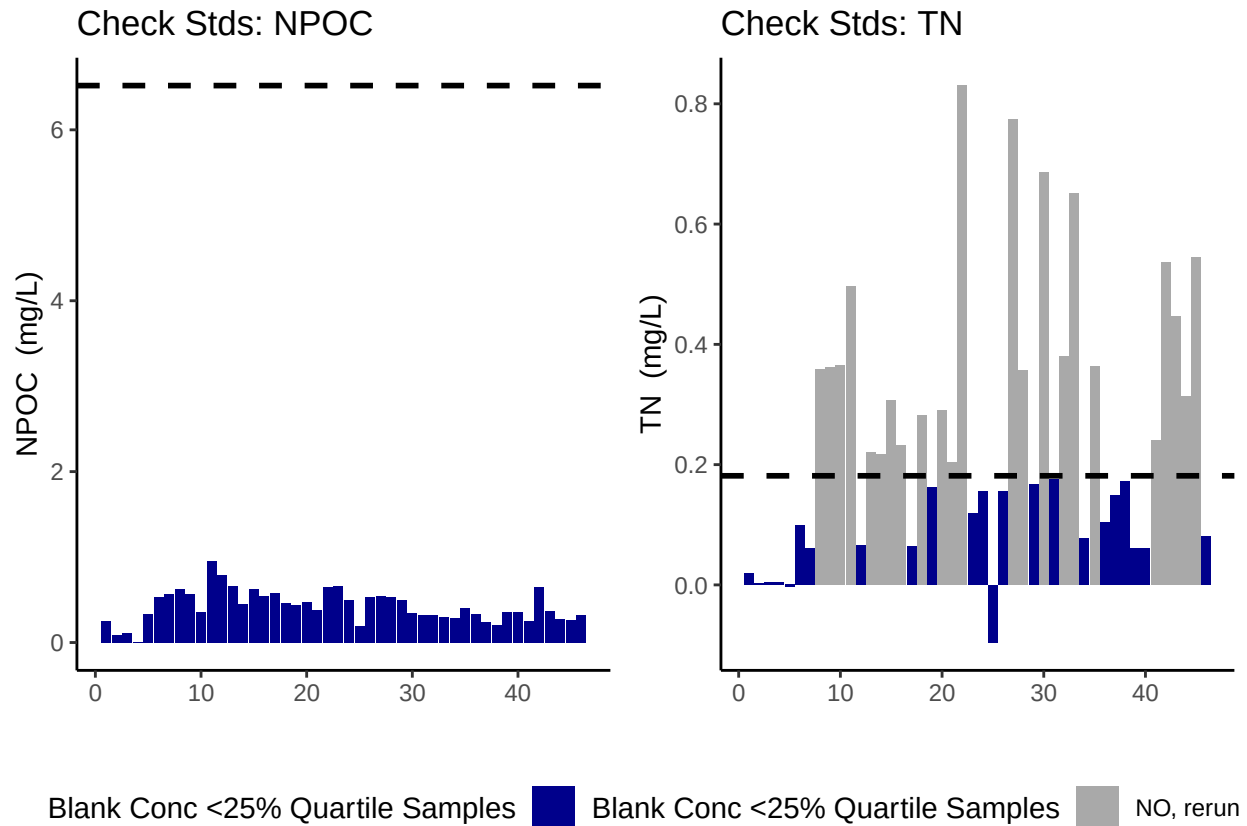
```
## Assess Blanks
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



```
## carbon blanks:
```

```
## [1] 0.4249826
```

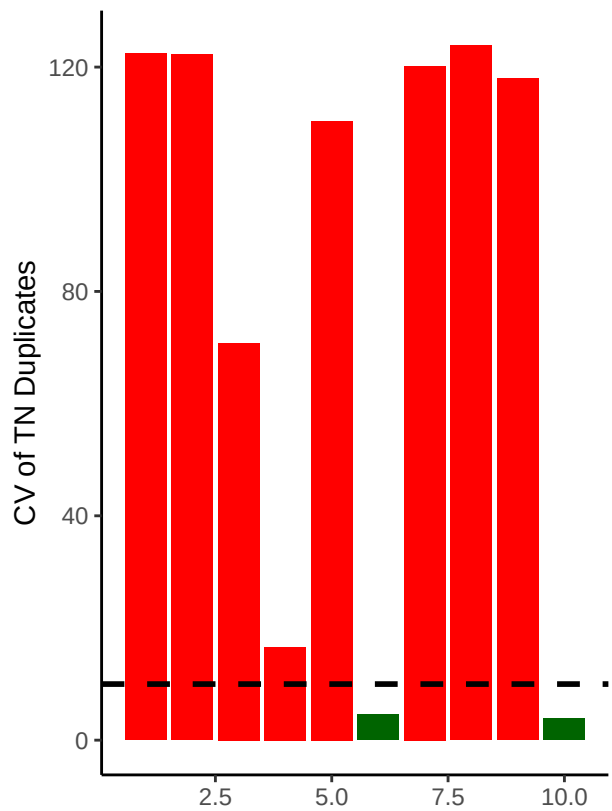
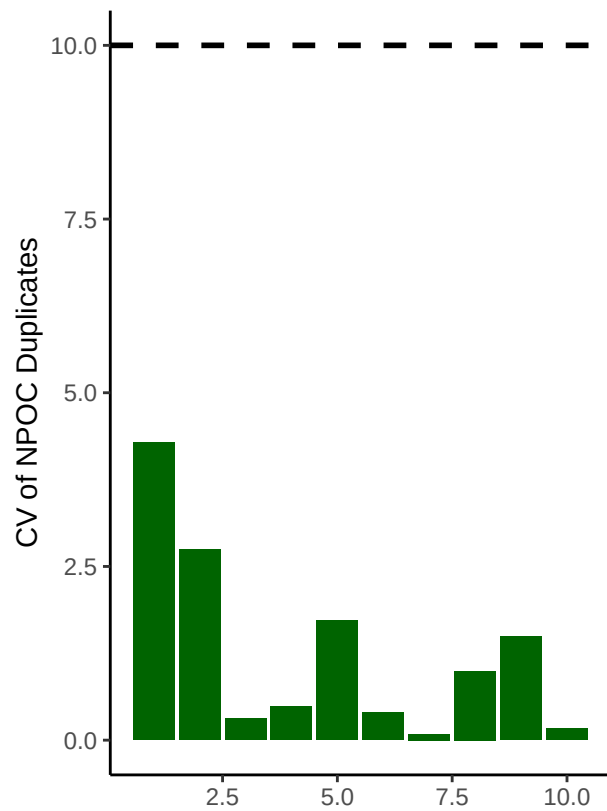
```
## nitrogen blanks:
```

```
## [1] 0.245943
```

0.6 Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

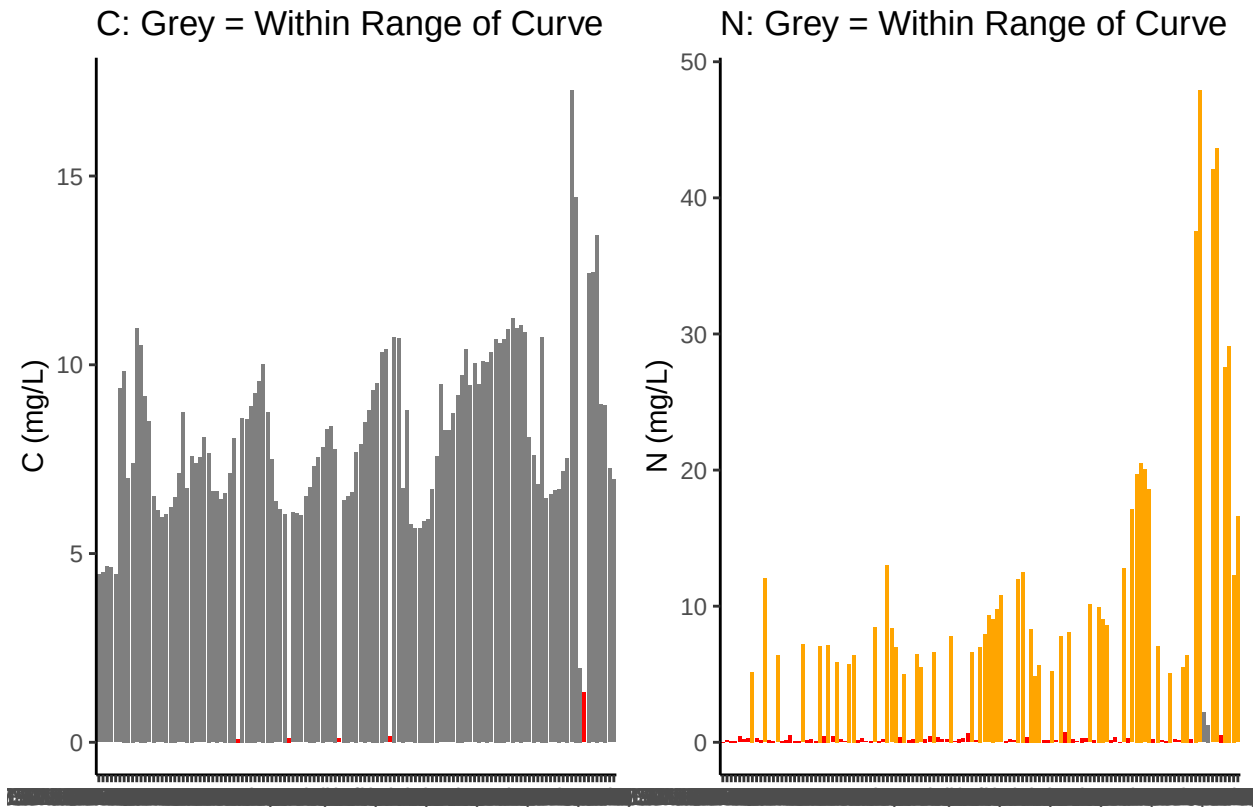


```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] "<60% of Nitrogen Duplicates have a CB <10% - REASSESS"
```

0.7 Sample Flagging - Are samples Within the range of the curve?

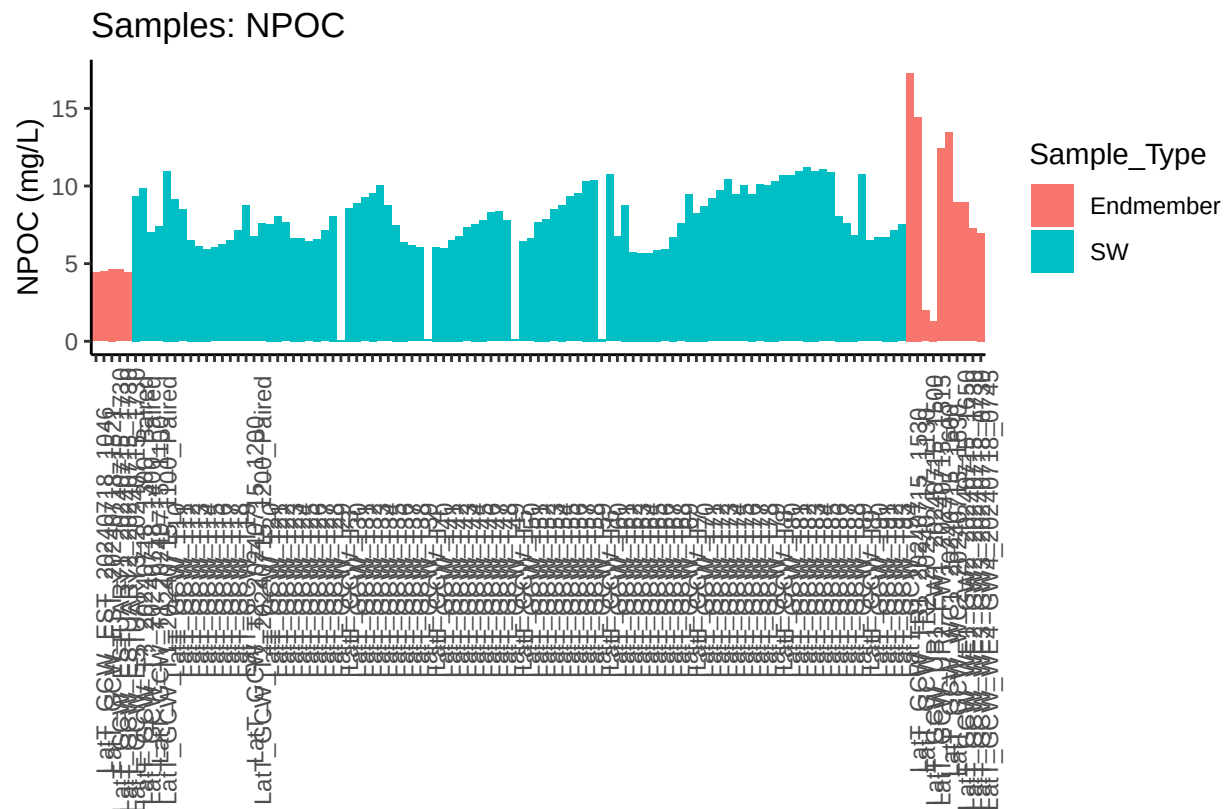
```
## Sample Flagging
```

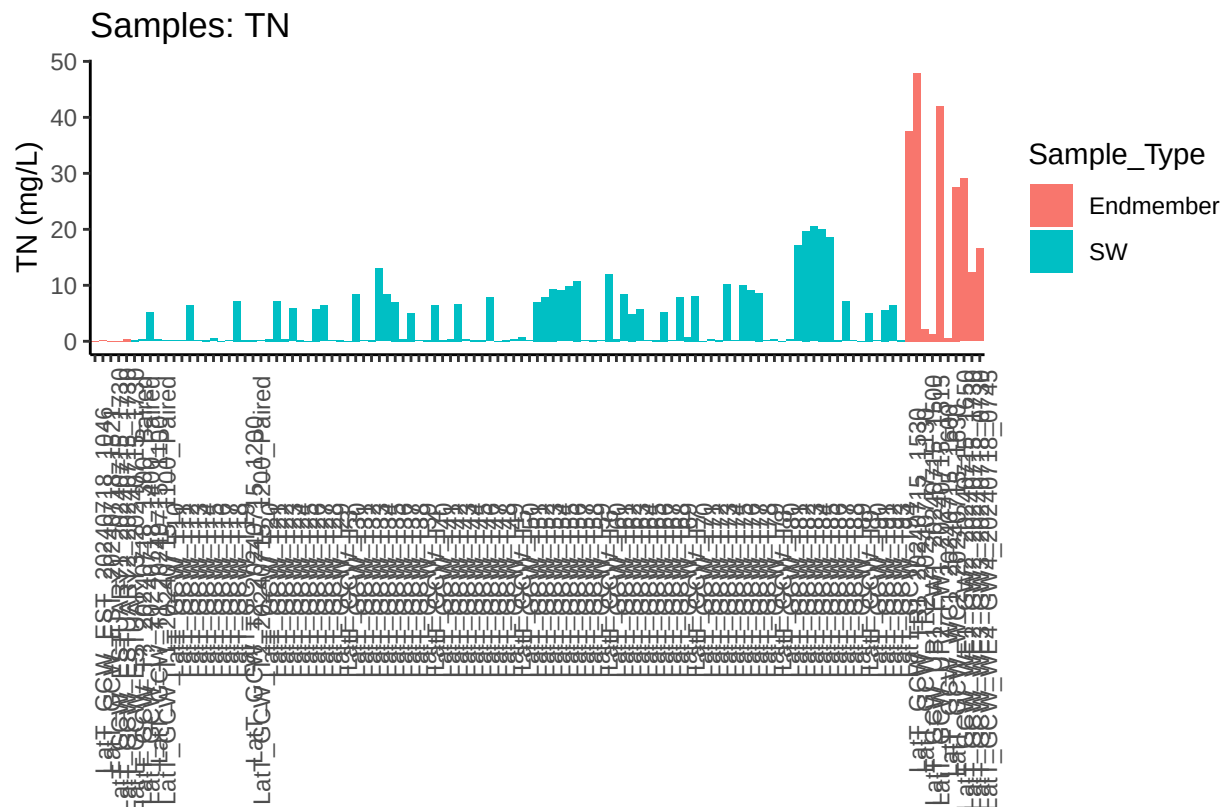


0.8 Visualize Data by Plot

```
## Visualize Data
```

```
## Warning in rbind(c("LatT", "GCW", "ESTUARY1", "20240715", "1730"), c("LatT", :  
## number of columns of result is not a multiple of vector length (arg 1)
```





0.9 Need to make TN measurements NA because bad run

0.10 Convert data from mg/L to uMoles/L

0.11 Export Processed Data

Export Processed Data

```
##          Project Region Experiment Site_Code Sample_Type Sample_ID
## 1 COMPASS: Synoptic      CB      LatT      GCW      Endmember ESTUARY1
## 2 COMPASS: Synoptic      CB      LatT      GCW      Endmember ESTUARY2
## 3 COMPASS: Synoptic      CB      LatT      GCW      Endmember ESTUARY3
## 4 COMPASS: Synoptic      CB      LatT      GCW      Endmember  UP1-FW1
## 5 COMPASS: Synoptic      CB      LatT      GCW      Endmember  UP2-FW1
## 6 COMPASS: Synoptic      CB      LatT      GCW      Endmember    TR1
## Sample_Date Time Paired          sample_name npoc_mgL  npoc_uM
## 1 20240715 1730 <NA> LatT_GCW_ESTUARY1_20240715_1730 4.660 387.9777
## 2 20240715 1730 <NA> LatT_GCW_ESTUARY2_20240715_1730 4.634 385.8130
## 3 20240715 1730 <NA> LatT_GCW_ESTUARY3_20240715_1730 4.443 369.9109
## 4 20240715 1500 <NA> LatT_GCW_UP1-FW1_20240715_1500 1.968 163.8498
## 5 20240715 1515 <NA> LatT_GCW_UP2-FW1_20240715_1515 1.318 109.7327
## 6 20240715 1530 <NA> LatT_GCW_TR1_20240715_1530 17.270 1437.8486
##          npoc_flag tdn_mgL tdn_uM
## 1                      NA      NA
```

```

## 2                NA      NA
## 3                NA      NA
## 4                NA      NA
## 5 blank is> 25% of sample value    NA      NA
## 6                NA      NA
##
##                                     tdn_flag
## 1 TN r2 low; TN blanks out of range; TN dups out of range; blank is > 25% of sample value
## 2 TN r2 low; TN blanks out of range; TN dups out of range; blank is > 25% of sample value
## 3 TN r2 low; TN blanks out of range; TN dups out of range; blank is > 25% of sample value
## 4                TN r2 low; TN blanks out of range; TN dups out of range
## 5                TN r2 low; TN blanks out of range; TN dups out of range
## 6                TN r2 low; TN blanks out of range; TN dups out of range; value above cal curve
##      Analysis_runtime      Run_notes
## 1 7/19/2024 3:31:23 AM The TN data is unusable. TN analyzer broke during run.
## 2 7/19/2024 3:55:34 AM The TN data is unusable. TN analyzer broke during run.
## 3 7/19/2024 4:18:46 AM The TN data is unusable. TN analyzer broke during run.
## 4 7/19/2024 4:41:26 AM The TN data is unusable. TN analyzer broke during run.
## 5 7/19/2024 5:03:55 AM The TN data is unusable. TN analyzer broke during run.
## 6 7/19/2024 5:33:14 AM The TN data is unusable. TN analyzer broke during run.

#end

```